



## **Diploma in Biomedical Engineering**

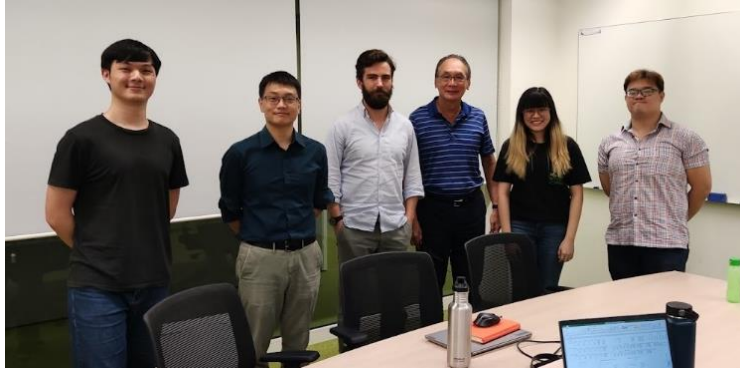
Student Internship Programme Final Report

Chin Fu Jie

1906886E

24 January 2022

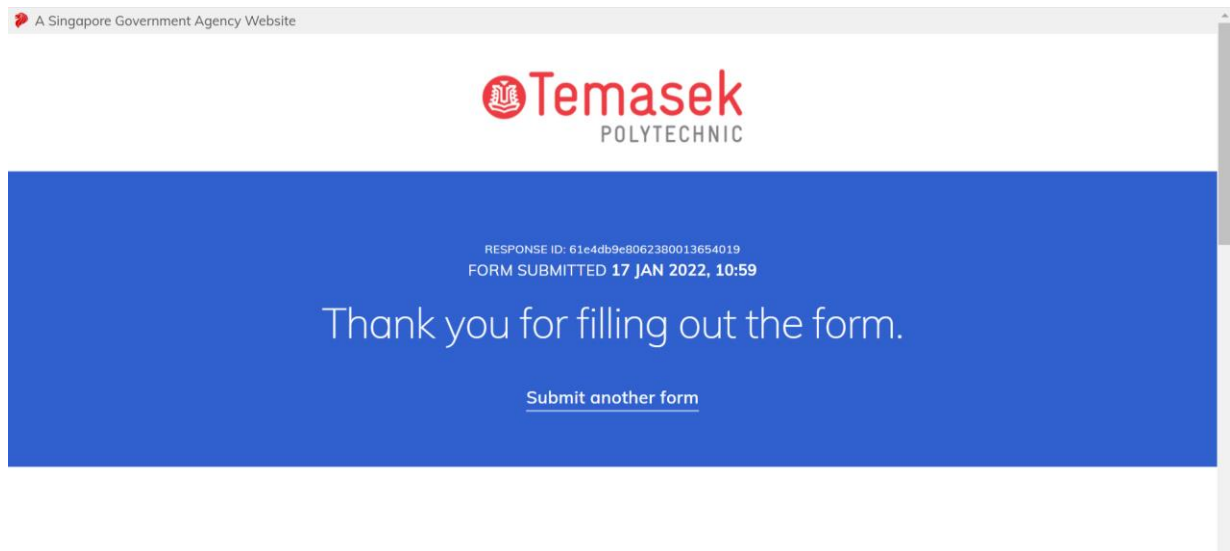
## ACKNOWLEDGMENTS



The author wishes to thank the research team and advisors of CellWave Technologies for their exceptional kindness and assistance: Mr. Robert Fu, Chief Executive Officer; Dr. Richard O'Rourke, Chief Technology Officer; Dr. Ye Ai, Chief Scientific Advisor; Ms. Pamela Chin, Product Engineer; Mr. Ralph Lin, Scientist. The author is also grateful to my Liaison Officer (TP) Mr. Qian Xi Jun, Lecturer, ENG/BME, for his support and advice.

### SIP experience survey

Chin Fu Jie E-survey acknowledgement screen:



### Learning Outcomes Form

Each student has a Learning Outcomes form that guides the learning process for the duration of the internship. This form should be acknowledged- and signed- by the three parties involved in the learning process, namely the student, the Liaison Officer and the Organisation Supervisor.

Students should insert their Learning Outcomes form here, as part of the submission of the Final Report. If the Final Report is submitted electronically, students can paste a screenshot of the form here.

**Temasek Polytechnic School of Engineering**  
**Diploma in Biomedical Engineering**  
**Student Internship Programme (SIP) Learning Outcomes**

Student Name: Chin Fu Jie

Admission No: 1906886E

SIP Organisation: CellWave Technologies

SIP Start Date: **20-Sep-21**

SIP End Date: 20-Feb-22

**Tick (✓)** the relevant learning outcomes that the student is expected to meet through the internship.

The set of selected outcomes will serve as an understanding between the Organisation and Temasek Polytechnic. The tables below should record the mutually agreed learning outcomes between the two parties at an **Initial** meeting. The student should be briefed on the learning outcomes. If there are subsequent changes to the learning outcomes, updates should be recorded at the **Final** meeting. If there are no subsequent changes, the Liaison Officer (LO) will still sign off and tick "No Changes in Learning Outcomes" after the **Final** meeting.

| S/No. | Learning Outcomes (Generic Life Skills)    | Initial | Final |
|-------|--------------------------------------------|---------|-------|
|       |                                            | ✓       | ✓     |
| 1.    | Communicates clearly and effectively.      | ✓       |       |
| 2.    | Works well in teams.                       | ✓       |       |
| 3.    | Works out solutions to day-to-day issues.  | ✓       |       |
| 4.    | Shows time and task management abilities.  | ✓       |       |
| 5.    | Shows initiative and independent learning. | ✓       |       |

| S/No. | Learning Outcomes (Diploma-Specific)                                                                                                                                                                         | Initial | Final |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|
|       |                                                                                                                                                                                                              | ✓       | ✓     |
| 1.    | Applies knowledge in biomedical electronics to design, develop, and/or troubleshoot electronic circuits for monitoring and analysing physiological signals                                                   | ✓       |       |
| 2.    | Applies essentials of programming, microcontroller firmware development and/or information technology to address problems related to biomedical engineering                                                  | ✓       |       |
| 3.    | Possesses knowledge and ability to operate and interpret a wide variety of biomedical equipment, including monitoring, diagnosis and/or life support devices                                                 | ✓       |       |
| 4.    | Demonstrates competency in conducting and interpreting laboratory experiments and/or in handling clinical laboratory equipment                                                                               | ✓       |       |
| 5.    | Applies knowledge in fundamental biological processes of the human body at cellular and organisation levels, and/or chemical and/or physical principles to solve problems relating to biomedical engineering | ✓       |       |
| 6.    | Applies knowledge of medical device regulations and bioethics, and/or biomedical and/or laboratory safety standards in the work environment                                                                  | ✓       |       |

| S/No. | Other Learning Outcomes* expected of the student, if not listed above.<br>(This section should be completed by the LO in consultation with the SIP organisation Supervisor.) | Initial | Final |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|
|       |                                                                                                                                                                              | ✓       | ✓     |
| 1.    | Communicates with suppliers on intended purchase                                                                                                                             | ✓       |       |
| 2.    | Product improvement via R&D                                                                                                                                                  | ✓       |       |

\*E.g. Manufacturing and quality improvement, Data analytics for product and customer needs, etc.

|                                               |
|-----------------------------------------------|
| <b>Remarks</b> (e.g. reasons for any changes) |
|-----------------------------------------------|

Learning  
Outcomes:  
(Initial)

Pamelia   
Name and Signature  
SIP Company Supervisor  
Date: 27-Oct-21

Qian Xi Jun   
Name and Signature  
Liaison Officer  
Date: 27-Oct-21

Chin Fu Jie   
Name and Signature  
Student  
Date: 27-Oct-21

Learning  
Outcomes:  
(Final)

\_\_\_\_\_  
Name and Signature  
SIP Company Supervisor  
Date: [Click here to enter  
a date.](#)

\_\_\_\_\_  
Name and Signature  
Liaison Officer  
Date: [Click here to enter  
a date.](#)

\_\_\_\_\_  
Name and Signature  
Student  
Date: [Click here to enter  
a date.](#)

☒ No Changes in Learning Outcomes (*SIP Company Supervisor's signature not required*)

## Mentorship Record Form

Each student should insert their Mentorship Record Form here.

### **SIP MENTORSHIP**

#### **OVERVIEW**

Mentorship refers to a personal developmental relationship where an experienced industry practitioner (mentor) provides professional and industry guidance to a student (mentee).

A Mentor takes on a long term view to provide guidance, support, encouragement and inspiration to the student through sharing knowledge and experience relevant to work, career and/or professional development. The intent is to develop the student's capabilities through a one-on-one relationship. **The Mentor could either be the organisation Supervisor or a senior staff assigned by the SIP organisation.**

Through the mentorship relationship, the student is kept aware of job and career growth opportunities in the industry/sectors for which they are being trained. The student will be encouraged to reflect on what he or she hopes to accomplish at internship and beyond, as well as the pathways to get there.

The student must be fully aware that it is a privilege to have a mentor while on internship as industry professionals are spending their time mentoring them. As such, students should initiate and maintain contact with their mentors to coordinate the meet ups.

It is recommended for the mentorship focus meeting to take place **at least 3 times** during the SIP i.e. initial period, during and at the end of the SIP.

Proposed topics for discussion at meeting could include, but are not restricted to the following:

- a) Knowledge, experiences and career journeys.
- b) Industry insights, developments and trends.
- c) Demands of the various career options.
- d) Personal development, career and/or professional development.

To fully reap the benefits of the mentorship, students should keep a record of the meetings, reflect on the discussions and record the action plans using the template in this document.

## SIP Mentorship Record

|              |                        |          |             |
|--------------|------------------------|----------|-------------|
| Student Name | Chin Fu Jie            | Admin No | 1906886E    |
| Diploma      | Biomedical Engineering | School   | Engineering |

|                                                                                                                                      |                                       |                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Mentor's Details</b><br><br>Name: <u>Pamelia Chin</u><br><br>Designation: <u>Ms.</u><br><br>Company: <u>Cellwave Technologies</u> |                                       | <b>Mentor Same as Supervisor? Yes / No</b><br><br><b>Mentor from same SIP Organisation? Yes / No</b><br><br><b>Remarks</b> |
| <b>Mentoring Session #1</b>                                                                                                          | Date: 16/11/2021<br><br>Time: 10:30am | <b>Other details</b>                                                                                                       |
| <b>Mentoring Session #2</b>                                                                                                          | Date: 08/12/2021<br><br>Time: 10:30am | <b>Other details</b>                                                                                                       |
| <b>Mentoring Session #3</b>                                                                                                          | Date: 06/01/2022<br><br>Time: 10:30am | <b>Other details</b>                                                                                                       |

*Note: Student may add additional session(s) and provide other details such as venue, whether it is a group session etc.*

| Areas Discussed                            | Student Reflections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Knowledge, experiences, and career journey | <p>My supervisor shared her studies and career journey with me, and it was very inspiring to me. Even though she had to continue her studies in a course not of her choice in polytechnic, she still tried her best and advanced her education to a local university, graduating in her desired degree with honours. She had also attempted to enter the workforce during the peak of the COVID-19 pandemic. Before being offered a full-time placement, she used traineeships to start her career journey in CellWave Technologies.</p> <p>I was fortunate enough for another co-worker to share with me his career journey as well. Interestingly, he had made a mid-career switch from his previous company to the current one while still utilising his knowledge in microbiology in CellWave Technologies research and development.</p> <p>Furthermore, as the company I was in specialises in the microfluidics area of research, I gained much knowledge of the industry. I learned that this</p> |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>research field is relatively new, and it has only been well defined in the past 20 years. It combines various well-known fields, including Photolithography and Integrated Circuits (IC) Technology. However, even though the research field is relatively new, its possible application in the Medical Industry causes it to remain a widely researched area.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Industry insights, developments, and trends | <p>I have learnt more about the situation of fresh graduates when entering the workforce, especially in the new normal of the COVID-19 pandemic. As my supervisor only graduated from university a few years ago, she shared what it was like for an engineering graduate to find a job. During the early stages of the pandemic, many companies had to resort to retrenchment, and they were also not willing to hire many new full-time employees. Therefore, my supervisor and her classmates had to apply for various traineeships. From there, she was accepted as a trainee and was subsequently offered a full-time placement due to her work quality and efficiency.</p> <p>In addition, another colleague had also shared about the growing food science industry due to numerous issues that had occurred recently.</p> <p>The first issue was that the global COVID-19 pandemic had caused some concern with Singapore's food security. This is especially true when Malaysia implemented the 'Movement Control Order', which restricts the movement of goods and people across our borders. With more than 90% of Singapore's food being imported from other countries, people started to panic and bought in excessive amounts, which caused supermarket shelves to be empty for a while. However, our food security was not at much risk since we import food from many suppliers worldwide. But the food science industry continues to be essential as the COVID-19 pandemic did bring about new food suppliers. Therefore, the need to test these fresh foods and ensure their safety for our people is more prevalent now than ever.</p> <p>Another issue that has contributed significantly to the food science industry recently is the increase in awareness of the negative impacts of the meat industry on our environment. Global livestock production accounts for seven gigatonnes of Carbon Dioxide every year, which contributes significantly to climate change and global warming, with beef being the most significant contributor. However, the demand for these meats will not decrease considerably as meat is a staple in most cuisines and daily meals. Therefore,</p> |



|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <p>people start to look for more environmentally friendly alternatives that still taste like real meat.</p> <p>One widely popularised and invested idea is lab-grown meat, where a small amount of meat tissue is harvested and grown in a lab. This alternative aims to serve meat more sustainably without the carbon emissions of herds of livestock while still ensuring that the meat comes from the actual source.</p> <p>Another exciting alternative to livestock is meat produced from the air. The process behind it is through a 'closed-loop carbon cycle', which has been used to feed astronauts on long space missions. Carbon dioxide in the air is captured by microorganisms called hydrogenotrophs and combined with other components to form protein.</p> <p>Therefore, with innovations to combat the ongoing fight against climate change, the food science industry continues to grow with more investors.</p> |
| Demands of the various career options                 | <p>I had learnt that the Research and Development Industry does encourage a higher level of education. However, a related degree is usually enough for an R&amp;D company in a similar field. R&amp;D is about learning along the way, and those who pursue their masters and PHD will have more experience after writing their various reports and thesis papers.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Personal development, career/professional Development | <p>My supervisor was kind to share her advice and various tips she had used personally throughout her education and career journey. She told me it was essential to analyse the situation well in anything I do and not just work hard, but to work smart. This way, I would maximise my time and increase my efficiency in anything I do, be it schoolwork or in the workplace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

*Note: Student may add additional area(s).*

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## **ABSTRACT**

This report records the work done and reflections during the period of the Student Internship Programme at CellWave Technologies from 20 September 2021 to 20 February 2022. CellWave Technologies' product development calls for vigorous research and development from a day-to-day basis. For these tests to take place, intricate components and devices are fabricated in large quantities. With current procedures being heavily reliant on manual procedures and prone to human error, which causes several devices to be rejected that wastes time and resources. The alignment of a laser beam to a specific area on the device's cartridge is another delicate procedure that is performed by hand. Checking the position of the cartridge is done manually and may cause deviations of a few micrometres that could potentially skew results, even if it may seem spot on by observation.

The task assigned to the student intern include producing solutions to the aforementioned problems, while assisting the company in other tasks and experiments throughout the product development. Several ideas were successfully tested and produced, such as the Automatic Cartridge Alignment Program and the 3D printed guide for forming holes within the PDMS during fabrication.

# 1. INTRODUCTION

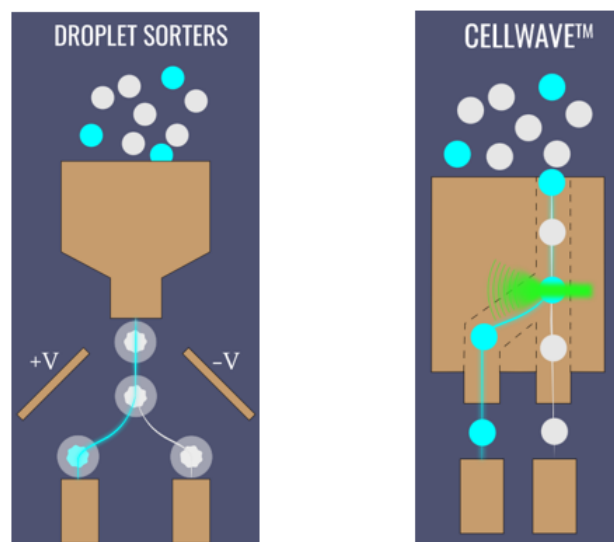
## 1.1 Background

CellWave Technologies Pte Ltd, founded in 2019, is a Research and Development company currently based in the Bio-Engineering & Allied Technologies Laboratory (BEATs Laboratory) of the Singapore University of Technology and Design (SUTD).



Figure 1: CellWave Technologies Logo [1]

CellWave Technologies was established on a newly discovered method for cell sorting. Cell sorting is necessary for essential scientific fields, including the medical sector, where patient samples must be cleared of unwanted and dead cells before analysis can start on the isolated targeted cells [1]. This is especially true with cancer cell detection, as medical experts have expressed interest in CellWave Technologies' ability to isolate and sort MCF-7 cells, a type of human breast cancer.



Figures 2: FACS (left) and CellWave Acoustic cell sorting (right) [1]

The current standard of Fluorescence-activated cell sorting (FACS) utilises electrically charged plates to deflect droplets of a sample into specific containers. However, this process causes extreme stress to the cells and leads to the deaths of upwards of 50% of them within the sorted output [1]. CellWave Technologies reconceptualises the cell sorting process and relies on acoustic waves within a microfluidic cartridge to isolate the intended cells (*Figure 3*).

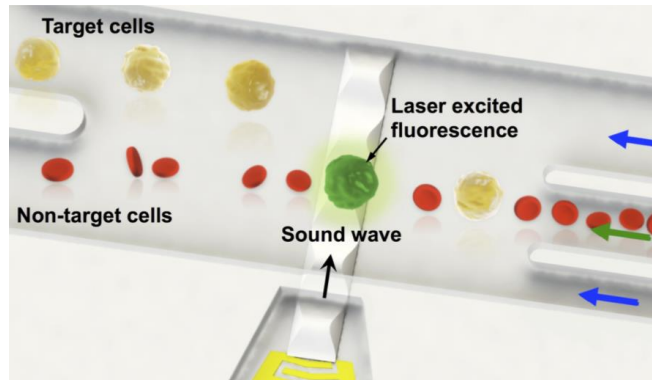


Figure 3: CellWave Technologies cell sorting with acoustic waves [1]

This process has been proven to be effective and gentler on the cell sample, retaining more than 95% of the sample's viability. This process can be used to remove dead cells by staining them with a suitable stain. Stained particles will be detected and removed, while the live cells will pass through the microfluidic channels undisturbed [1]. All the functions and processes occur within the POC prototype, as seen in *Figure 4* below.



Figure 4: Proof of concept (POC) prototype

With scientific papers supporting the technology behind the device and a functional Alpha and Beta prototype, the company had been shortlisted into the top-20 cohort of MedTech Innovator APAC 2020 [2].



Figure 5: MedTech Innovator APAC [2]

## 1.2 Attachment Period and Work Environment

During the 22-week attachment from 20 September 2021 to 20 February 2022, one member was tasked to identify possible solutions to the fabrication and testing processes that were heavily manpower reliant and prone to human error and produce a solution for them. The intern from Temasek Polytechnic also assists the researchers in laboratory tests and liaises with relevant materials and product suppliers when necessary. In early December 2021, a successful demonstration of the Automatic Cartridge Alignment Program was shown to the research team, including the supervisor, followed by a write-up and a video showing its functionality for subsequent investor pitching.



Figure 6: BEATs Lab Workbenches

The current location where CellWave Technologies conducts its research is in the Bio-Engineering & Allied Technologies Laboratory (BEATs Laboratory), located on the 7<sup>th</sup> floor of building 2 of Singapore University of Technology and Design (SUTD). The facility is split into two sections, the office and the laboratory—an external storeroom beside the main laboratory stores laboratory and safety equipment.



Figure 7: Laboratory Access Card

An access card is issued for the student intern to enter the SUTD BEATs laboratories (*Figure 7*). The working duration last from Monday to Friday every week, 7 hours with an hour lunch break daily. The reporting time is 9 am, and the clock out time is 5 pm. The time for a lunch break is flexible and will depend on the experiments and tasks for that day. The attire required is casual with long pants and covered shoes. The co-workers were very helpful and friendly, especially when teaching me the necessary information needed at the laboratory and the company. Degree holders pursuing their doctorates were also present at the laboratory, conducting experiments for their research and thesis papers. They were also very sociable and willing to help when requested. The laboratory is well maintained, with clear tabletops and organised shelves. Protective equipment such as disposable nitrile gloves and lab coats were provided for use in the laboratory.

## 2. OBJECTIVES

### 2.1 Internship Objectives

This internship aimed to assist CellWave Technologies in various tasks and projects to further advance and improve their medical device's feasibility and experiments workflow while gaining insights into the field of microfluidics.

Some of these tasks and projects include:

- Identifying a solution to align a cartridge in the prototype device automatically and consistently within only a few microns in deviation;
- Come up with a solution to improve the device fabrication workflow and reduce human error;
- Fabricate Polydimethylsiloxane (PDMS) devices for fluidics, particle detection and sorting tests;
- Contact material and product suppliers and request quotations and samples utilised in our product testing;
- Assist researchers with laboratory experiments and conduct pre-experiment and post-experiment preparations and checks.

### 2.2 Personal Objectives

On top of the internship objectives, the student intern planned to achieve:

- Expand knowledge on the Research & Development industry;
- Learn more on the science and engineering technicalities behind various laboratory equipment and microfluidics;
- Apply and refine soft skills and hard skills developed during the time in polytechnic;
- Experience and learn of work ethics and professionalism in the workforce;
- Discover the various education and career paths from experienced co-workers.



### 3. WORK EXPERIENCES

#### 3.1 Cartridge Alignment Program

The current Beta Proof of Concept (POC) Prototype operates by first inserting a cartridge into the device. A cell sample is then loaded up into the machine and passes through the cartridge for cell sorting to occur. Within the cartridge, small microfluidic channels direct the flow of the sample, along with a liquid buffer, towards the cell sorting region. Two significant operations occur in this cell sorting area of the cartridge (*Figure 8*).

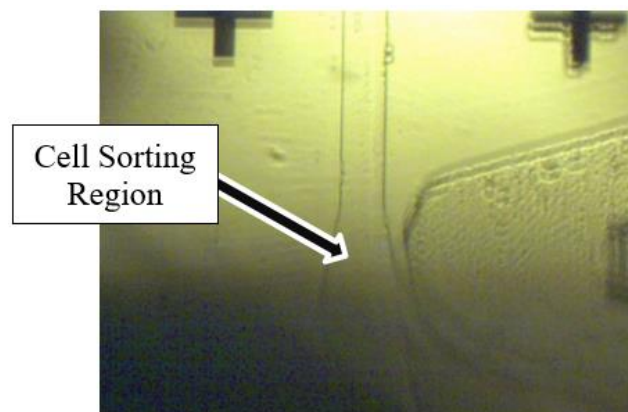


Figure 8: Cell sorting region of PDMS device

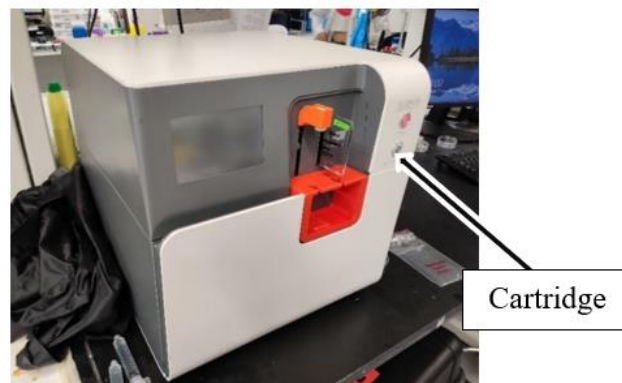


Figure 9: POC prototype with Cartridge inserted

First, the sample flow will directly pass through a laser beam. Similar to a standard flow cytometer, the laser will produce a fluorescence effect within the pre-stained and labelled cell sample that is subsequently detected by the device. Particles that pass through the laser will also scatter the laser light, with larger particles causing a larger scatter of light. With both the fluorescence and forward scatter parameters, the device can accurately define the particle or cell passing through the microfluidic channel at that point in time.

Next, the device uses the information received by the detectors and decides whether to activate an acoustic wave to push the cell into another path, depending on if the cell is of the right fluorescent level and size determined by the forward scatter of the laser light.

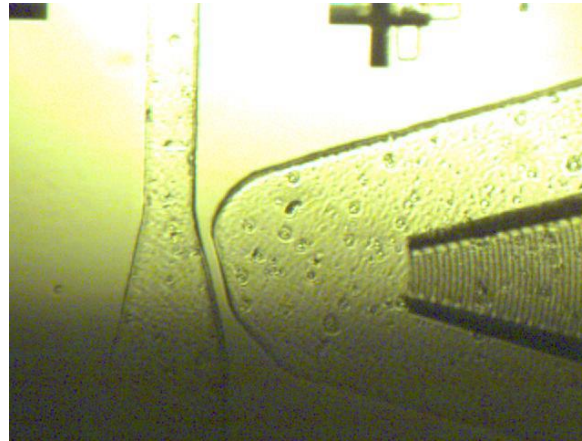


Figure 10: Acoustic wave generator on the right

These two processes continue for each particle and cell in the sample until it has completed sorting it. However, this process requires the laser to be accurately aligned to the sample flow within the cartridge, only allowing a deviation of a micrometre or two. If the laser is not aligned, signals from the detectors can change, and the device would not be able to function appropriately to perform cell sorting.

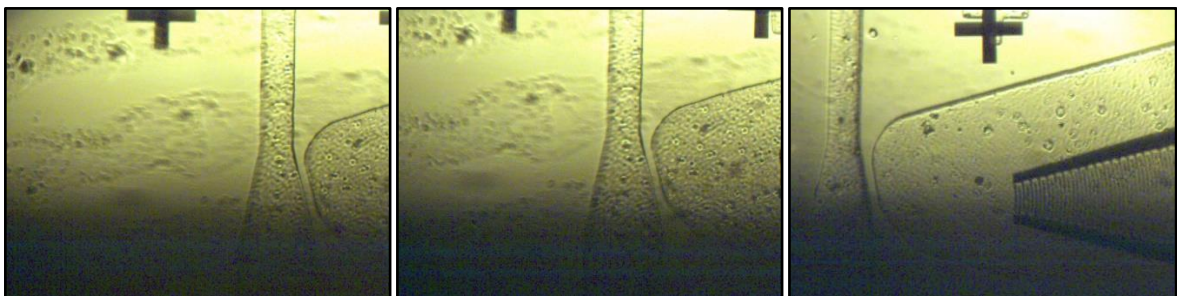


Figure 11: Various cartridge starting positions

Traditionally, the research team manually aligned the cartridge whenever it was inserted into the device. Even with the cartridge holder consistently in the same place, the cartridge would always deviate further than a few micrometres from the optimal laser position (*Figure 11*). Thus, accurately aligning the cartridge for testing is prone to human error and time consuming to rectify.

Thus, the first task assigned to the student intern is to suggest and produce a viable solution to automatically align the cartridge down to the micrometre whenever it is inserted into the Beta POC Prototype device.

### 3.1.1 Learning Python programming language

Development on an Automatic Cartridge Alignment Program started in early October. Python has been chosen as the coding language as much existing software on the POC Prototype device have also been developed in Python. Thus, future integration of the Cartridge Alignment Program into the current graphical user interface (GUI) will be straightforward.

The student intern did not have much knowledge of the Python programming language. However, he had prior experience with similar programming languages such as C and Arduino programming languages. Thus, the intern familiarised himself with the Python programming language and relevant libraries for the hardware involved in the operation.

### 3.1.2 Basler Pylon Camera



Figure 12: Basler Cameras [3]

The POC prototype is equipped with a Basler Pylon Camera to view the microfluidic sorting area when the cartridge is inserted. Images and videos can be captured from the camera, either by the Pylon Viewer Application or commands in the Python script. However, the camera can only be used by one application or program at a time, not allowing for simultaneous viewing of the camera view and the program's execution.

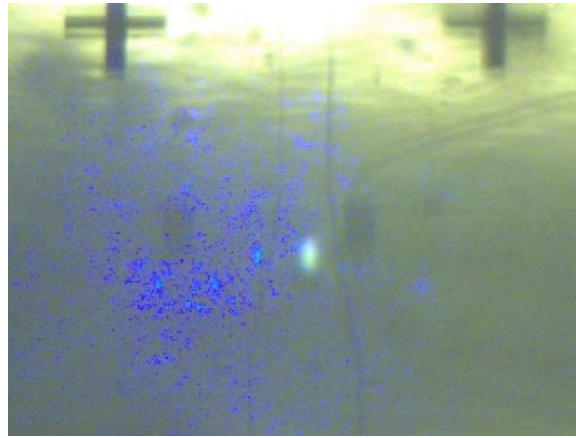


Figure 13: Ideal Cartridge position aligned to the laser spot

An ideal cartridge position is determined and recorded from the camera as a reference for the Cartridge Alignment Program from prior experiments (*Figure 13*). As the position of the laser spot will always remain the same in respect to the camera's field of view, the program will only need to shift the cartridge into the same ideal position consistently.

Conveniently, there are alignment markers on the microfluidic chip previously used to fabricate the PDMS device attached to the cartridge. These cross-shaped markers are then used to determine the exact position of the cartridge on the X and Y axes and subsequently adjusted to the correct position by moving the cartridge.

### 3.1.3 Newport Picomotors

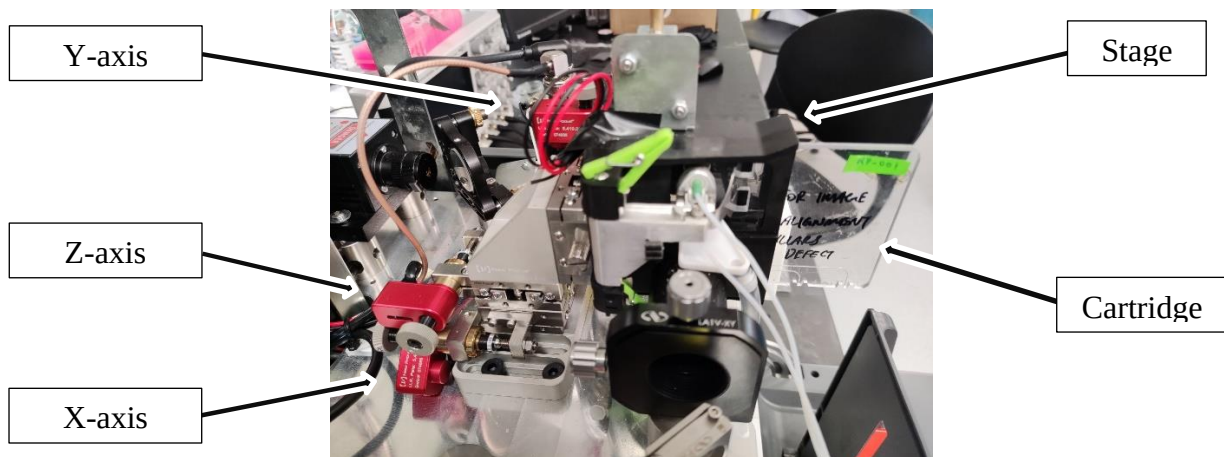


Figure 14: X, Y, and Z Picomotors

The cartridge stage is moved by three Newport Picomotors, which can make minimal adjustments to the cartridge position. The picomotors are each capable of only moving the cartridge on one axis. Thus, the three picomotors covers all three X, Y and Z axes of movement. The picomotor shifting the cartridge in the Z-axis determines the focus of the camera view, while the picomotors moving the cartridge in the X and Y axes adjust it in the respective planes.

The picomotors are connected to a controller, where they can be controlled by the Newport Picomotor Application or through commands sent through a USB connection. However, the picomotors can only be controlled by one application or program at a time.

After the program determines the distance to move, commands are sent to the respective picomotors through serial communication to make the necessary adjustments. Sample picomotor movement commands taken from Newport's user manual and sample program script can be seen in *Table 1* below:

Table 1: Picomotor Movement Commands

|              |                                             |
|--------------|---------------------------------------------|
| Command Line | oUSB.Query(strDeviceKey, "1PR+{}", strBldr) |
|--------------|---------------------------------------------|

|                                                                            |
|----------------------------------------------------------------------------|
| Text in the inverted commas can be modified to move in various directions: |
| "1PR+{x}": move cartridge upwards by "x" amount                            |
| "1PR-{x}": move cartridge downwards by "x" amount                          |
| "2PR+{x}": move cartridge towards the left by "x" amount                   |
| "2PR-{x}": move cartridge towards the right by "x" amount                  |
| "3PR+{x}": move cartridge towards the lens by "x" amount                   |
| "3PR-{x}": move cartridge away from the lens by "x" amount                 |

After some testing, it was realised that the commands sent for movement in one direction of the picomotors did not correspond to the distance travelled if the same command was sent in the opposite direction. Therefore, tests were conducted to create look-up tables for calibrating the relation between the commands sent and the actual distance covered by the picomotors, as seen in *Table 2* below.

Table 2: Newport Picomotor distance codes to estimated distance travelled

| Upwards                                      |                  | Downwards                                    |                  |
|----------------------------------------------|------------------|----------------------------------------------|------------------|
| Command                                      | y-shift (pixels) | Command                                      | y-shift (pixels) |
| 1PR+500                                      | -10              | 1PR-500                                      | +8               |
| 1PR+1000                                     | -20              | 1PR-1000                                     | +15              |
| 1PR+5000                                     | -93              | 1PR-5000                                     | +76              |
| 1PR+6750                                     | -123             | 1PR-8000                                     | +122             |
| $y\text{-shift} = \frac{\text{Command}}{54}$ |                  | $y\text{-shift} = \frac{\text{Command}}{64}$ |                  |

| Left                                         |                  | Right                                        |                  |
|----------------------------------------------|------------------|----------------------------------------------|------------------|
| Command                                      | x-shift (pixels) | Command                                      | x-shift (pixels) |
| 2PR+500                                      | -7               | 2PR-500                                      | +5               |
| 2PR+1000                                     | -13              | 2PR-1000                                     | +11              |
| 2PR+5000                                     | -68              | 2PR-5000                                     | +52              |
| 2PR+11000                                    | -146             | 2PR-14000                                    | +146             |
| $x\text{-shift} = \frac{\text{Command}}{73}$ |                  | $x\text{-shift} = \frac{\text{Command}}{93}$ |                  |

Delays were also necessary to pause the program for the duration of the picomotors operation. Thus, tests were done to create another set of look-up tables for the relation between the commands sent and the time taken for the picomotors to cover the pre-determined distance (*Table 3*).

Table 3: Newport Picomotor distance codes to the estimated time taken

| Upwards                                 |                      | Downwards                               |                      |
|-----------------------------------------|----------------------|-----------------------------------------|----------------------|
| Command                                 | Time Taken (seconds) | Command                                 | Time Taken (seconds) |
| 1PR+6750                                | 3.5                  | 1PR-8000                                | 4.1                  |
| 1PR+13500                               | 6.9                  | 1PR-16000                               | 8                    |
| 1PR+27000                               | 13.55                | 1PR-32000                               | 16                   |
| Time Taken (s) = $\frac{Command}{1910}$ |                      | Time Taken (s) = $\frac{Command}{1940}$ |                      |

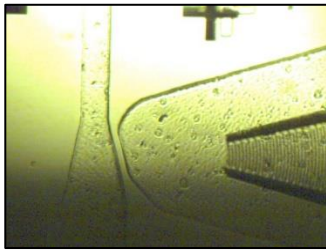
  

| Left                                    |                      | Right                                   |                      |
|-----------------------------------------|----------------------|-----------------------------------------|----------------------|
| Command                                 | Time Taken (seconds) | Command                                 | Time Taken (seconds) |
| 2PR+5500                                | 3                    | 2PR-700                                 | 3.8                  |
| 2PR+11000                               | 5.5                  | 2PR-14000                               | 7.2                  |
| 2PR+22000                               | 10.5                 | 2PR-28000                               | 14                   |
| Time Taken (s) = $\frac{Command}{1990}$ |                      | Time Taken (s) = $\frac{Command}{1930}$ |                      |



### 3.1.4 Cartridge Alignment Program Workflow

1



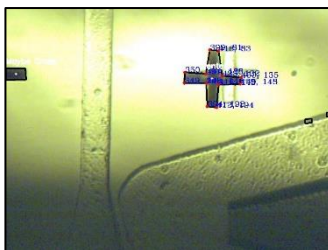
The current image of the starting position is captured by the camera when the program is started.

2



The program identifies cross markers/potential marker edges and chooses the most likely candidate.

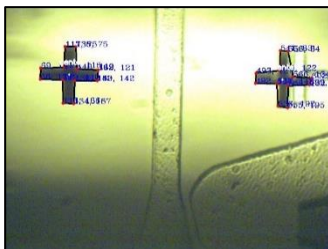
3



Picomotors adjust the cartridge position to better view the potential marker and check if it is indeed a cross marker.

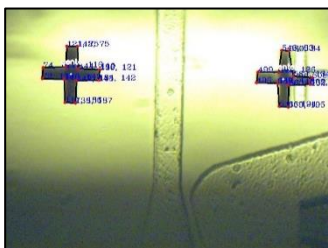
The cartridge will shift to the next likely marker if it is not.

4



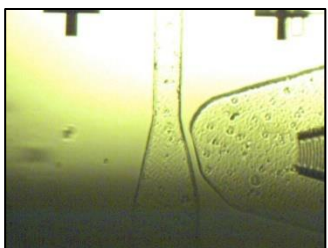
Once the cross marker is identified, picomotors adjust the cartridge for both cross markers to come into the camera's field of view.

5



Minor adjustments are made such that the x-coordinates of both cross markers are sufficiently aligned to the predetermined values from the Ideal Cartridge Position.

6

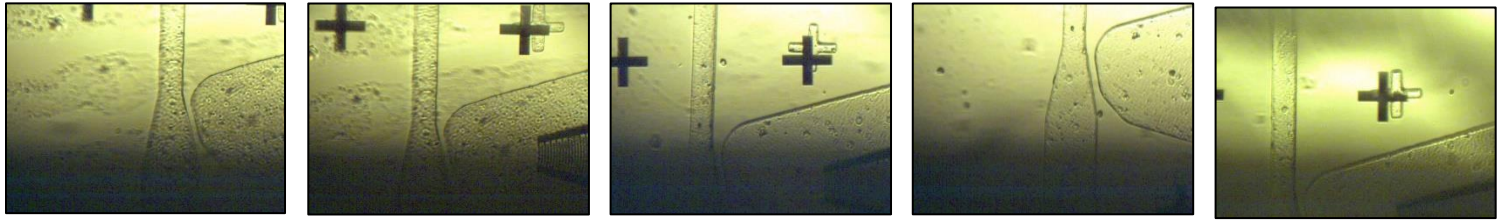


The distance required for the cartridge to be shifted upwards is calculated, and the picomotors make a final adjustment in the y-axis.



### 3.1.5 Results and Accuracy

#### Initial Positions



#### Final Positions

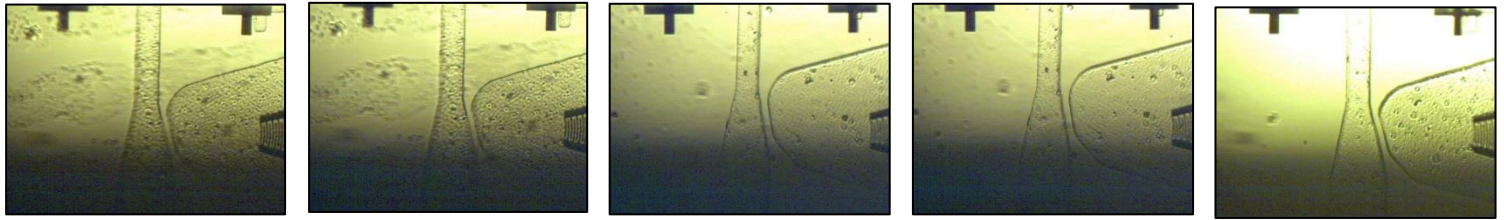


Figure 15: Various starting (top row) and end positions (bottom row)

As seen in the figures above, even with the various degrees of deviation in the cartridge's starting position, the Cartridge Alignment Program can still reposition the cartridge consistently.

Table 4: Data analysis using ImageJ software

|                    | Pixel Coordinates |    |              |    | Cross distance from origin (Microns) |          |              |          |
|--------------------|-------------------|----|--------------|----|--------------------------------------|----------|--------------|----------|
|                    | Left Marker       |    | Right Marker |    | Left Marker                          |          | Right Marker |          |
|                    | X                 | Y  | X            | Y  | X                                    | Y        | X            | Y        |
| Expected Values    | 128               | 61 | 542          | 61 | 183.2177                             | 87.31468 | 775.8124     | 87.31468 |
| Test Results       | 121               | 64 | 545          | 58 | 173.198                              | 91.60885 | 780.1066     | 83.02052 |
|                    | 125               | 67 | 548          | 59 | 178.9235                             | 95.90301 | 784.4008     | 84.45191 |
|                    | 126               | 69 | 548          | 66 | 180.3549                             | 98.76579 | 784.4008     | 94.47162 |
|                    | 119               | 63 | 542          | 58 | 170.3352                             | 90.17746 | 775.8124     | 83.02052 |
|                    | 117               | 61 | 540          | 68 | 167.4724                             | 87.31468 | 772.9496     | 97.3344  |
|                    | 121               | 59 | 545          | 64 | 173.198                              | 84.45191 | 780.1066     | 91.60885 |
|                    | 122               | 56 | 545          | 64 | 174.6294                             | 80.15774 | 780.1066     | 91.60885 |
|                    | 120               | 56 | 544          | 59 | 171.7666                             | 80.15774 | 778.6752     | 84.45191 |
|                    | 121               | 51 | 545          | 56 | 173.198                              | 73.0008  | 780.1066     | 80.15774 |
|                    | 120               | 54 | 544          | 61 | 171.7666                             | 77.29496 | 778.6752     | 87.31468 |
|                    | 123               | 59 | 547          | 63 | 176.0608                             | 84.45191 | 782.9694     | 90.17746 |
|                    | 120               | 59 | 545          | 63 | 171.7666                             | 84.45191 | 780.1066     | 90.17746 |
|                    | 126               | 61 | 550          | 65 | 180.3549                             | 87.31468 | 787.2635     | 93.04024 |
| Average            |                   |    |              |    | 174.0788                             | 85.77319 | 780.4369     | 88.52586 |
| Standard Deviation |                   |    |              |    | 3.899864                             | 7.262598 | 3.748755     | 5.188861 |

The ImageJ software used for processing and analysing scientific images is used to examine the accuracy of the Cartridge Alignment Program. As seen in *Table 4* above, the program achieves precision with a Standard Deviation of 3.7-3.9 micrometres on the X-axis and 5.2-7.3 micrometres on the Y-axis.

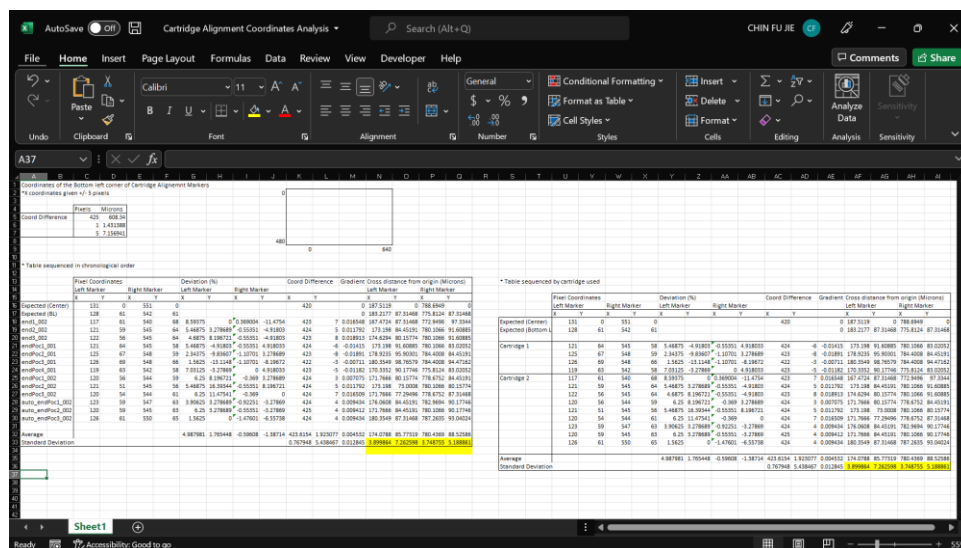


Figure 16: Analysis and calculations on Microsoft Excel

The X-axis standard deviation can be reduced as the current program allows for a +/- 5-pixel deviation (7 microns). Therefore, this range of pixel values can be lowered to increase the program's accuracy. For the Y-axis, the obscuration of the top half of the marker reduces the program's accuracy in this plane. Thus, a dedicated marker placed closer to the sorting area will be added in future designs to increase accuracy by rectifying the cartridge's Y-axis position. Currently, a multiple shift procedure will be implemented in the coming days, instead of a single shift step in the Y-axis, to help verify the distance travelled in the Y-axis and detect deviations in the X-axis when moving the cartridge towards the final position.

### 3.2 PDMS Device Fabrication

Polydimethylsiloxane (PDMS) devices are commonly used for various biomaterial applications, such as contact lenses. It has been popularised in the scientific community as the material used to fabricate different types of devices for studying neural stem cell behaviour [4]. As PDMS is classified as a silicone, it is used in soft lithography to produce designs on it after it cures and hardens into a solid [5].

In CellWave Technologies, the PDMS is imprinted with microfluidic channels using soft lithography on a predesigned SU-8 mould and subsequently bonded to a microfluidic chip after plasma treatment of the surfaces.

### 3.2.1 SU-8 Mould Fabrication using Photolithography

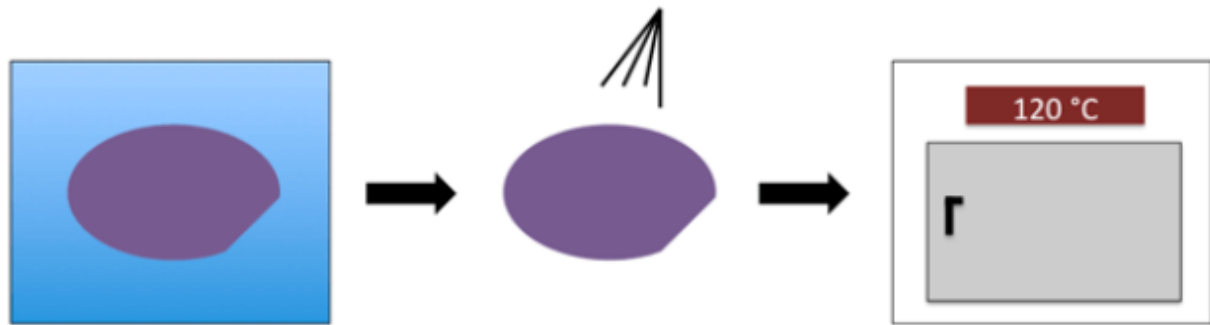


Figure 17: Wafer cleaning and heating [5]

The master mould is fabricated entirely in the cleanroom. First, a blank silicon wafer is cleaned with Acetone before being heated up at 120°C in an oven. This step will allow the SU-8 to better attach to the substrate later on [5].

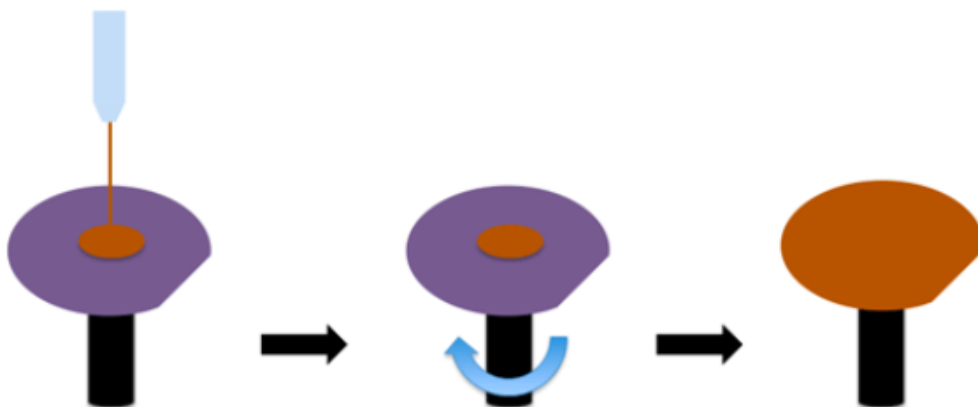


Figure 18: SU-8 Photoresist on the Spin Coater [5]

Next, a spin coater is used to produce the SU-8 photoresist layer. A predetermined volume of SU-8 photoresist is placed on the middle of the wafer of the rotated wafer. The rotation velocity, acceleration, as well as viscosity of the SU-8 will determine the thickness of the SU-8 layer.

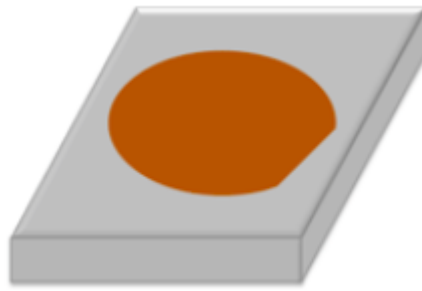


Figure 19: Soft Bake on the hot plate [5]

The wafer piece is then transferred to a hot plate for soft baking, where it is heated up and cooled down at a slow rate to prevent stresses within the SU-8 photoresist. This step evaporates the solvent and hardens the SU-8.

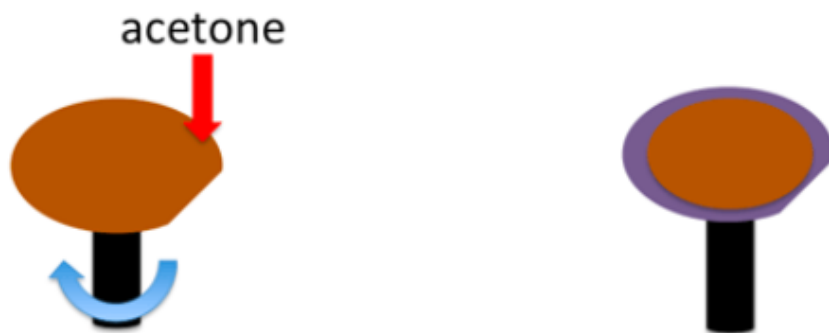


Figure 20: Edge removal with Acetone [5]

The edge of the photoresist is then removed, either manually or by applying Acetone at it when spun at high velocities. This is because the edges are several microns thicker than the rest of the wafer and thus need to be removed.

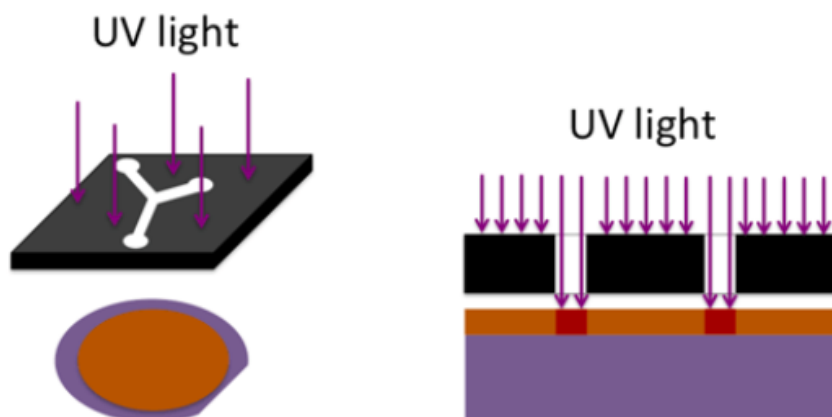


Figure 21: UV exposure with photomask design [5]

Now, the SU-8 is exposed to UV light through a photomask with the desired designs to be imprinted onto its surface. As SU-8 is a negative photoresist, the UV exposure hardens the SU-8 with the photomask designs. The photomask design used contains the microfluidic channels to be transferred to the PDMS later on.

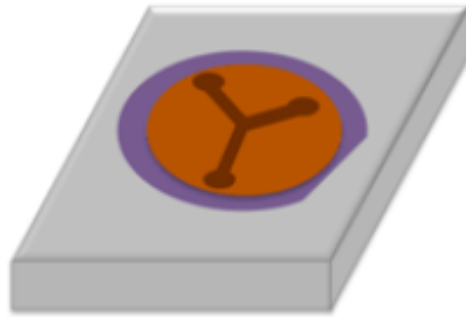


Figure 22: Exposure Bake on the hot plate [5]

The wafer piece is baked once again on a hot plate to provide energy to photoactive components of the SU-8, continuing the hardening process of the surfaces exposed to the UV light. The heating and cooling of the wafer have to be gradual to avoid internal stresses of the SU-8.

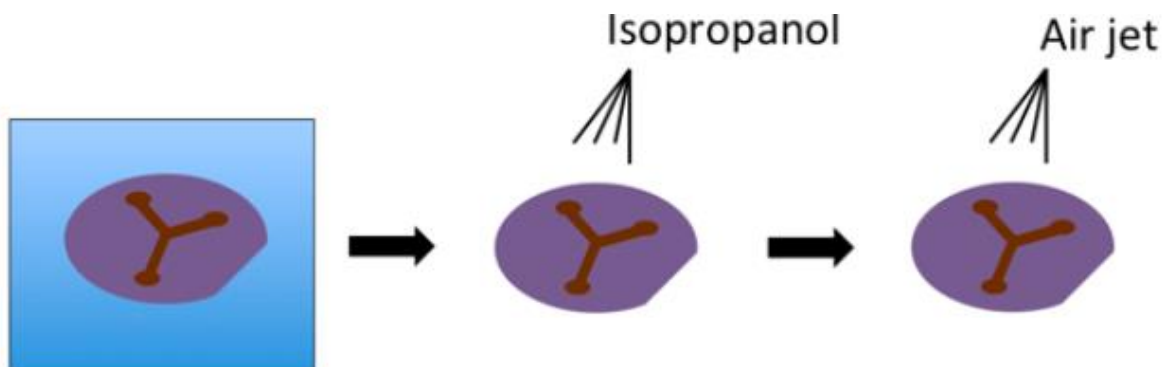


Figure 23: Development of SU-8 designs [5]

The SU-8 wafer is then diluted in a solvent, where the areas which are not hardened will be dissolved and removed. Now the designs will be visible on the wafer. The SU-8 is then developed with Ethyl-lactate or Di-acetone alcohol before it is cleaned with Isopropanol and then dried.

Lastly, the mould is baked one last time to strengthen the SU-8. This hard bake heats the wafer to high temperatures to quell internal stresses that may cause cracks and delamination. The

SU-8 mould is now complete, and a visual inspection of the surface using a surface profiler and microscope can be performed to verify its quality.

### 3.2.2 PDMS Piece Fabrication using Soft Lithography

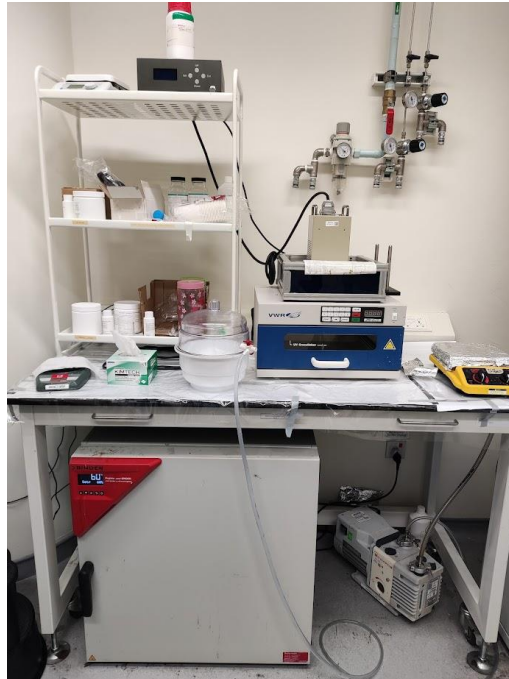


Figure 24: Microfluidic room and scientific instruments

With the SU-8 mould, soft lithography can be performed to transfer the microfluidic designs from the master mould onto the PDMS. First, the PDMS is prepared in a microfluidics room by mixing the two ingredients, an elastomer base and a curing agent. The mixture will be prepared in a one to ten ratio, 1-part curing agent and 10-parts elastomer base, which is measured by weight on a weighing scale. This ratio will produce a piece of PDMS with the optimal physical properties and biocompatibility [6].

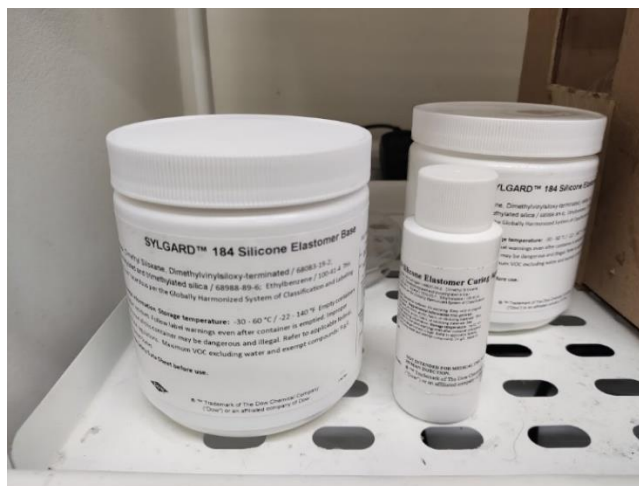


Figure 25: PDMS base and curing agent

The liquid PDMS is then mixed thoroughly before it is placed in a Vacuum chamber to remove any air bubbles that had been produced. After the liquid PDMS has been degassed, it is then transferred to a petri dish containing the SU-8 mould. The petri dish acts as a container to hold the liquid PDMS as it cures into a solid. The volume of liquid PDMS poured into the petri dish will determine the thickness of the PDMS piece. Thus, the volume of liquid PDMS used will vary depending on the type of PDMS piece needed for the experiment.

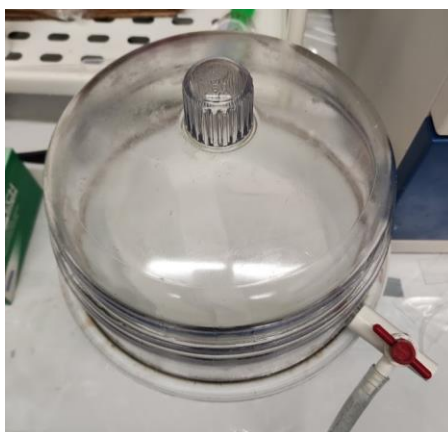


Figure 26: Vacuum chamber



Figure 27: Cured solid PDMS on SU-8 mould



Next, the petri dish is placed into the oven and heated to accelerate the curing process. After the PDMS has cured, it is cut and removed from the SU-8 mould, and the microfluidic channel designs will have transferred onto the PDMS piece. The SU-8 mould will then be cleaned and be ready to be reused.



Figure 28: Oven used for PDMS curing

The PDMS piece will be cut up into smaller individual pieces. Subsequently, a microfluidic PDMS puncher is used to punch a hole into specific areas of the PDMS, which is used for fluidic connection later [7].



Figure 29: Microfluidic PDMS puncher

A magnifying glass is used to assist in aligning the microfluidic PDMS puncher to the microfluidic channel during this process and verified using a microscope to spot any misalignment in the punch (*Figure 30*).



Please refer to Appendix C, Figures 19 and 20 for more information on quality checks on the PDMS punch.



Figure 30: PDMS QC checks using a microscope

After a batch of PDMS has been punched, they are gathered in a beaker of Deionised (DI) water before being placed in an ultrasonic cleaner to remove any particles and dust (*Figure 31*). As the microfluidic channels are a few micrometres in diameter, any debris within the channel will compromise test results and sample collections. Therefore, multiple washing steps will take place in the fabrication of the PDMS device.



Figure 31: Ultrasonic Cleaner

The cleaned PDMS pieces are then dried using compressed air from an air gun before being taped up and stored in a petri dish to avoid contact with dust particles. Please refer to Appendix A for more information on lab equipment used.



Figure 32: Air gun for compressed air

### 3.2.3 Plasma Cleaning and Bonding

The next step would be to bond the PDMS piece to the acoustic chip through plasma bonding. The acoustic chip produces an acoustic wave at a specific area when a resonant radio frequency (RF) is applied to it from an RF generator.

Before undergoing plasma treatment, the PDMS piece will be washed with Isopropyl Alcohol (IPA), and the acoustic chip is washed with Acetone to remove any contaminants from their surfaces. After drying the pieces with compressed air, they are aligned on an alignment setup comprised of a camera connected to a screen, a stage that allows for manual adjustment, and the PDMS holder that brings the two pieces together when they are sufficiently aligned (*Figure 33*). Dedicated markers on the PDMS piece and acoustic chip are used to visualise their alignment before they are bonded together.

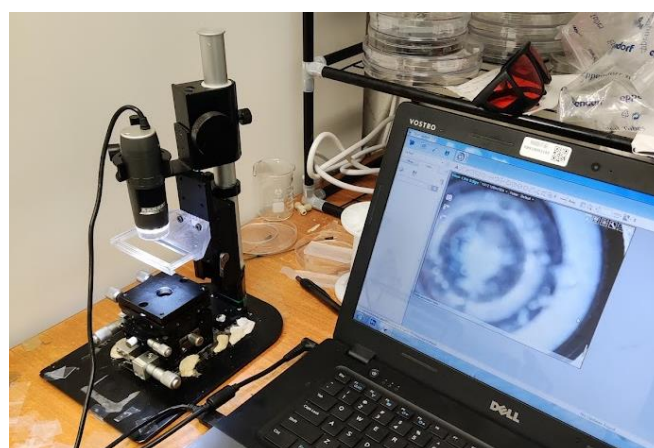


Figure 33: Alignment setup used for PDMS and acoustic chip bonding

The cleaned pieces are then placed in a plasma cleaner, where ionised gas can produce plasma in a vacuum environment (*Figure 34*) [8]. Thus, a vacuum is pulled into the chamber before the treatment begins. During the plasma treatment process, the pieces are exposed to the plasma for a few minutes to remove impurities and contaminants while activating the surfaces of the PDMS and acoustic chip for bonding.



Figure 34: Plasma cleaner used for plasma treatment

After plasma treatment, the pieces are quickly taken out of the plasma cleaner and placed back onto the alignment setup. There is only a short time frame for the bonding to occur before dust and other particles inhibit the bonding process. If necessary, a quick readjustment is made before the two pieces are brought together and compressed. The PDMS device is completed, and it is placed on a hot plate for an hour before it is ready for use (*Figure 35*).



Figure 35: Hot plate used for fabrication steps

### 3.2.4 Pre-formed Holes Prototype Design

As seen in the previous sections, the PDMS device fabrication uses many tedious manual steps and procedures that come with some degree of human error. Although the pieces and devices undergo quality control (QC) checks to prevent the use of faulty devices, the discarded pieces would have wasted time and resources. Therefore, removing some manual aspects to the fabrication process was assigned.

During the process of punching holes into the PDMS piece, the hole produced needs to be within a few micrometres of deviation to become a functional device. However, the process is done by hand with the microfluidic PDMS puncher, and its accuracy is verified by observation. Furthermore, several punches are made on the same PDMS piece, which increases the likelihood of a defective PDMS piece, as one misaligned punch causes the whole piece to be unusable.

A prototype was designed to overcome this issue by skipping the hole punching procedure altogether. By altering the process at which the liquid PDMS is cured, holes can be pre-formed into the PDMS as it is curing in the oven.

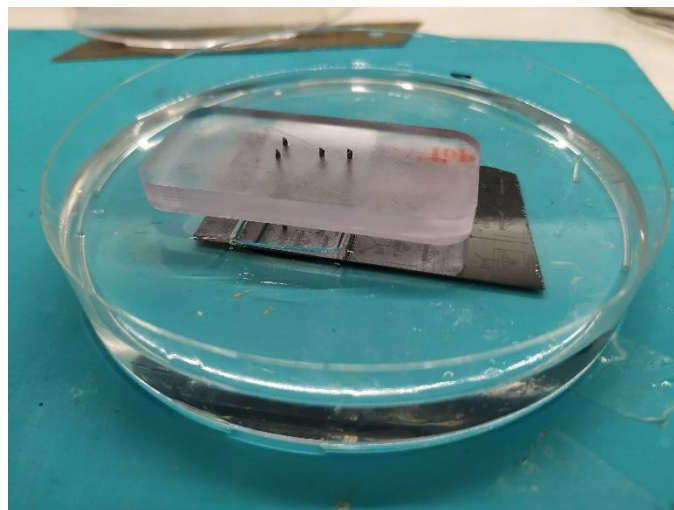


Figure 36: 3D Printed piece holding 0.83mm diameter needles

As seen in the figure above, a 3D printed piece holding needles of 0.83mm diameter is placed onto the SU-8 wafer. For the test, needles are aligned onto a single microfluidic channel of the SU-8 mould at the areas where holes need to be produced. PDMS is then poured into the petri dish and surrounds the needles before the setup is placed in the oven for the PDMS to cure.

Please refer to Appendix A, Figures 11 and 14 for more information on the 3D printer and equipment for UV curing.

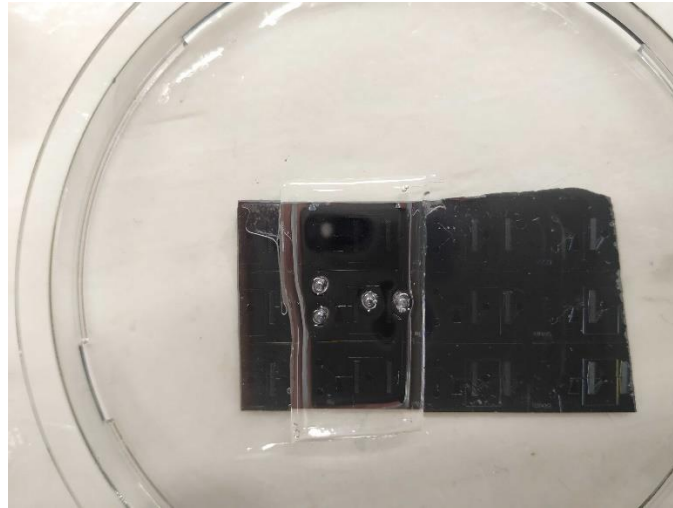


Figure 37: Cured PDMS with formed holes

After curing, the needles were removed, and the PDMS pieces were cut out. The holes were aligned to the desired locations, but bumps were observed at the areas at which the PDMS contacts the needles (*Figure 38*).

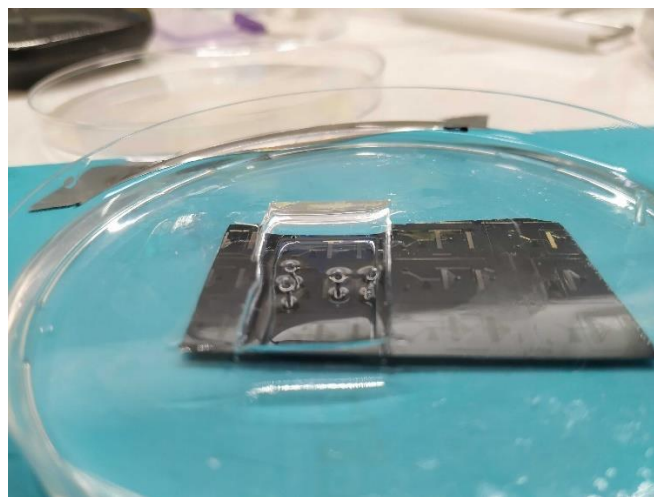


Figure 38: Bumps around the holes after curing

The bumps were evident and could potentially cause some laser light scattering and affect the PDMS device's performance. This problem is alleviated by performing a second degassing procedure with the whole 3D printed setup in the vacuum chamber.



Please refer to Appendix B for more information on the experiment procedures and result analysis.

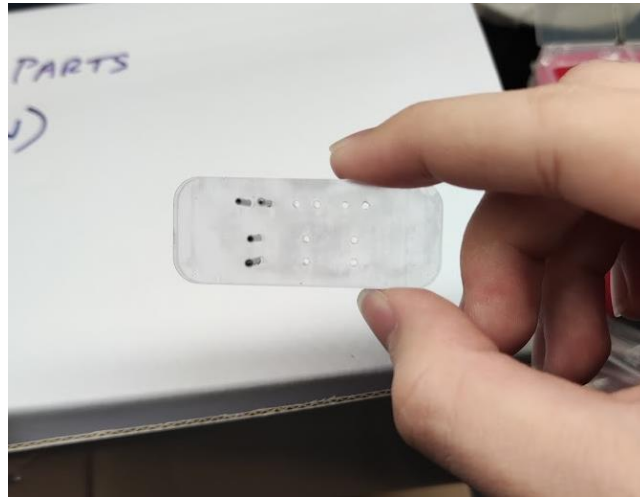


Figure 39: Modified 3D printed piece for multiple needles

The 3D printed prototype's design is altered to test the accuracy of aligning needles to multiple microfluidic channels in hopes of creating a set for the whole SU-8 master mould. The experiment succeeded, and the needles were aligned correctly to the microfluidic channels. Plans for expanding the scale to cover an entire SU-8 wafer is underway.

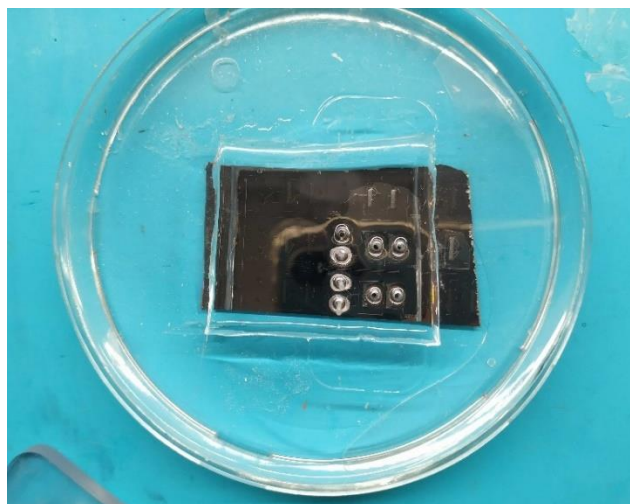


Figure 40: PDMS produced with two sets of holes

### 3.3 POC Prototype Experiments and Quality Control Checks

There are various steps to take before research and tests can be done on the POC prototype to ensure the components are working as intended. The same applies after the experiment, where the post-experiment clean-up is essential to keeping the cartridges and POC prototype functional and uncontaminated for subsequent use.

### 3.3.1 Cartridge and PDMS Device Setup

The cartridge and PDMS device are inspected for any sort of debris within its channels that may affect the fluidics' flow under an inverted microscope. If contaminants are observed, the cleaning liquid is flowed through it depending on the type of contaminant. This step is carried out before any experiment, and it is crucial after the experiment is completed. This is because the channels are guaranteed to contain some form of particles after they have been used (*Figure 41*). Thus, they need to be cleared before they can be used again.



Figure 41: Debris seen in microfluidic channels

If the contaminant is biological, such as cells, bleach is flushed through to lyse and clear them from the channels. If beads from previous experiments are observed, the Pluronic F-127 solution, which acts as a surfactant to remove them [9]. Regular DI water is then used to clear the remaining debris and cleaning solutions from the channels.

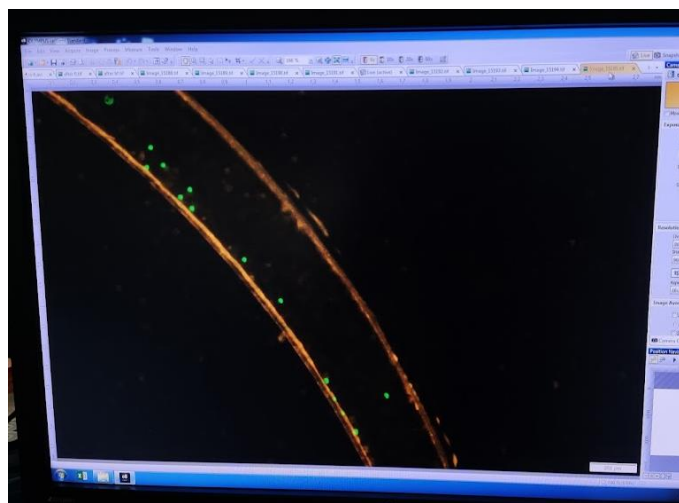
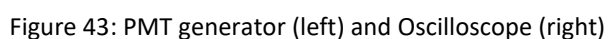


Figure 42: Fluorescent Beads in a microfluidic channel seen under the microscope

Please refer to Appendix C, Figures 22 and 23 for more information on the Quality Control checks of different post-experiment debris.

The POC prototype is switched on, along with other devices such as the Photomultiplier Tube (PMT) to generate the laser beam and the Oscilloscope to observe the output signals (*Figure 43*). Various programs such as the camera, XYZ stage controls, fluidics and voltage settings are set up during this time.



Some soldering is required to produce connectors for connecting the POC prototype to the PDMS device on the Cartridge.



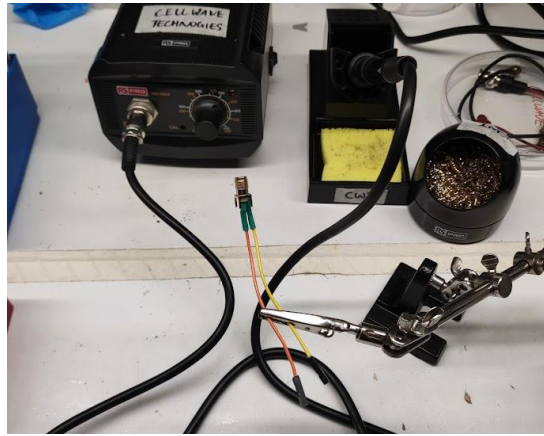


Figure 44: Soldering connector heads to wires

### 3.3.4 Impedance Analysis

The acoustic chips undergo QC checks for impedance using the Impedance Analyser device and the Trewmac Analyser software. If the waveform from the Impedance Analyser shows an expected reading, the acoustic chip will then be used on the cartridge.



Figure 45: Impedance analyser

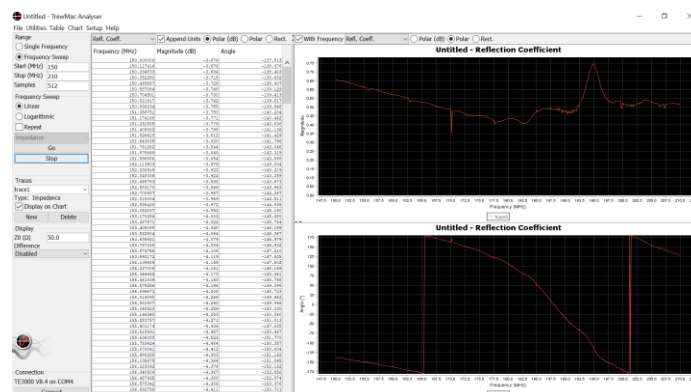


Figure 46: Trewmac Analyser software

### 3.3.5 Flushing POC System

After each experiment, there might be remnants of the chemicals or particles used that may remain in the POC prototype's fluidic system. Therefore, a standard cleaning procedure using a python GUI is used to clear out the system.

### 3.4 Request for Quotations

Optimal materials and equipment are essential to improve a device or make a workflow more efficient. These materials and products include desktop cutters, rubber sheets and biocompatible single sided pressure-sensitive adhesive (PSA) tape. Thus, several requests for quotations were sent out to notable suppliers on the availability and price for these products.

The prices of various desktop cutters were noted down and presented to the research team for feedback for the desktop cutters. The supplier was also willing to test if the material we wanted to cut was suitable for the product.

Samples for the rubber sheets were collected from the manufacturer and tested in the laboratory. Similarly, quotations and lead times for the PSA films were consolidated and reported to the research team. The best options were identified, purchased, and subsequently tested with commonly used chemicals such as bleach and DI water.



Figure 47: Rubber sheet samples acquired from manufacturer

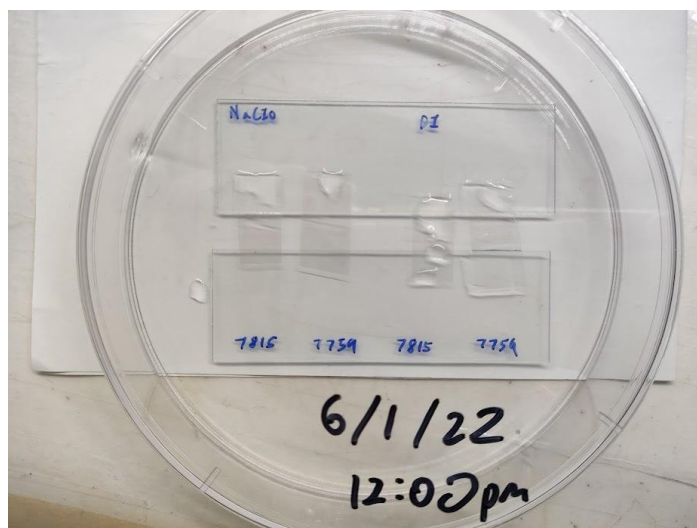


Figure 48: Chemical testing on product from different suppliers

## 4. INTERNSHIP REFLECTIONS

Over the course of the internship, I have gained various knowledge and skills, ranging from many hard skills and soft skills. I have been exposed to a very new environment, which has led me to learn new hard skills such as theoretical knowledge of microfluidics, cell culturing, and even learning how to request quotations. Existing hard skills were also utilised and improved upon during this internship, including soldering, programming, flow cytometry, 3D design and printing, and proficiency with Microsoft Office applications.

The same can be said for soft skills, as skills used in the workplace such as communication, problem-solving, team coordination, and a good work ethic are further built upon. Specific skills such as meeting presentation skills and being an active listener were also improved on.

### 4.1 REFLECTIONS ON GENERIC LIFE SKILLS ACQUIRED

Working in CellWave Technologies has taught me many things about the working world, especially within the Research and Development industry. Working as a team helped me build on my communication skills as information must be conveyed clearly and consistently within the team. The information can include keeping each other informed on test results, tasks to be done, inventorying of remaining supplies, among many others. Therefore, proper company communication helped me improve my work efficiency and team coordination.

Another generic life skill I have improved upon is problem-solving. This is because being tasked with solving various problems and brainstorming ways to improve workflows individually or as a team has helped me grow in this area. It has made me think outside of the box and refine my thinking process depending on the situation I am put in.

Having a good work ethic is another life skill I have learned in the workplace. A good work ethic would encompass qualities such as being dependable, productive, cooperative, and professional. Being disciplined to follow the rules and emphasising on producing quality work are more traits I have learned in CellWave Technologies.

With the company having meetings every other week, I have learned various meeting skills and insights. I had the chance to observe the different perspectives and views of the company members and how information should be conveyed to different types of people. People in charge of researching and producing results will be more inclined to want more time and resources to refine their process. In contrast, people who would deal with the company's finances will be drawn towards the product's attractiveness to the customer and investor perspectives. All in all, the many meetings I have been in, either online or conducted physically, helped me improve my presentation skills. I have learnt that I must be well prepared before the meeting and know what I want to say. I will also need to convey my thoughts clearly, either through easily understandable slides or through well-crafted speeches.

## 4.2 REFLECTIONS ON TECHNICAL KNOWLEDGE AND SKILLS ACQUIRED

During the start of the internship, I was given a Chemical Safety Awareness Training, Biosafety Training, and a Laboratory Induction to learn the dos and don'ts in the SUTD BEATs lab. The Chemical Safety Awareness Training helped me know the standard procedures to identify and handle certain chemicals. It has taught me always to check the safety data sheets of unfamiliar chemicals that I needed to use to prevent any chance of an accident. It had also taught me the steps to take for different types of emergencies.

As for the Biosafety Training, I have learned of different risk group classifications for biological agents. Risk group 1 has low to no individual and community risk, and risk group 4 has a high individual and community risk. The laboratories are also classified according to biosafety levels

that correspond to the degree of safety measures and the capability to handle dangerous agents. Similarly, biosafety level 1 will have lower safety measures in place to handle safer biological agents, while biosafety level 4 will have stringent safety measures to deal with dangerous pathogen units. Safety measures and disposal methods were also highlighted, with a strong emphasis on safety.

| Biosafety Training                          |                                         |                                                        |
|---------------------------------------------|-----------------------------------------|--------------------------------------------------------|
| 21                                          |                                         |                                                        |
| Relation of risk groups to biosafety levels |                                         |                                                        |
| Risk Group                                  | Biosafety Level                         | Laboratory Type                                        |
| 1                                           | Basic – Biosafety Level 1               | Basic teaching, research                               |
| 2                                           | Basic – Biosafety Level 2               | Primary health services; diagnostic services, research |
| 3                                           | Containment – Biosafety Level 3         | Special diagnostic services, research                  |
| 4                                           | Maximum Containment – Biosafety Level 4 | Dangerous pathogen units                               |

ENVIRONMENTAL, HEALTH & SAFETY

Figure 49: Biosafety Training example slide

A laboratory induction was conducted physically to familiarise me with the whereabouts and correct usage of equipment and substances in the laboratory. This was especially helpful as it helped me include equipment I had not seen in the lab before in my thinking process and workflow. Having familiarised myself with the standard procedures and safety equipment, I was better prepared to work in the laboratory.

I had also gained much knowledge of the biomedical applications of the microfluidic industry. With the relatively new field of research, microfluidics has gained popularity in its potential to create any kind of design needed that is well defined in PDMS and integrate electronics onto them. Developing the CellWave POC prototype has shown me how microfluidics can improve existing technology in the medical sector and potentially save and improve lives.

Even though I had not done cell culturing in the laboratory, I was fortunate enough to have passionate co-workers to show me how it is done correctly. The whole process of preparing the cells to grow is done in a biosafety cabinet.

Within it will contain the necessary equipment and chemicals for cell culturing. Since growing cells require an incredibly sterilised environment, every required object must be sterilised with

an ethanol spray before entering the biosafety cabinet. The biosafety cabinet itself comes with an automatic sterilisation process using UV light, and it is performed before and after each use [10].

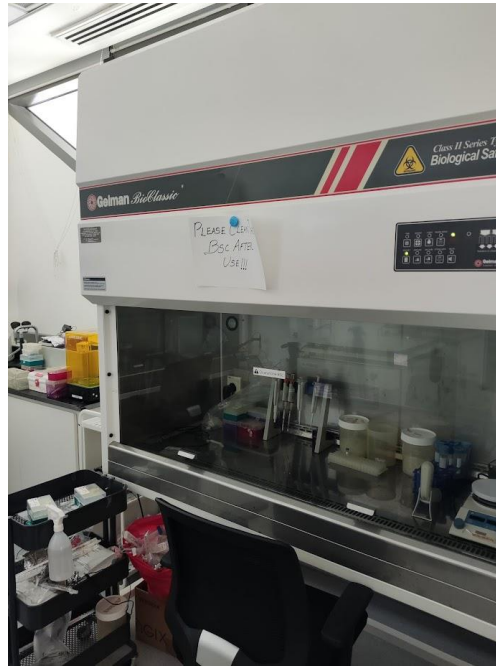


Figure 50: Biosafety Cabinet

In my case, I observed the cell culturing of the MCF-7 cell lines. They are a type of human breast cancer cells that will be grown for cell sorting tests on the POC prototype. The ones we had obtained were cryogenically preserved in liquid nitrogen a few years prior thus they were thawed in a warm water bath before cell culturing began. Next, the vial of cells is then centrifuged to check if there is enough required for cell culture. As the cells are denser than the surrounding liquid, they will settle at the bottom of the vial after centrifuging for easy inspection.

After that, the vial of cells is then placed into the biosafety cabinet along with the required equipment for cell culture. This equipment includes electronic pipettors, disposable pipettes, beakers, an ethanol spray, a tube rack with vials of cells and reagents, media, and cell culture flasks, all of which are sterilised by ethanol to avoid contamination of the cells (*Figure 51*).

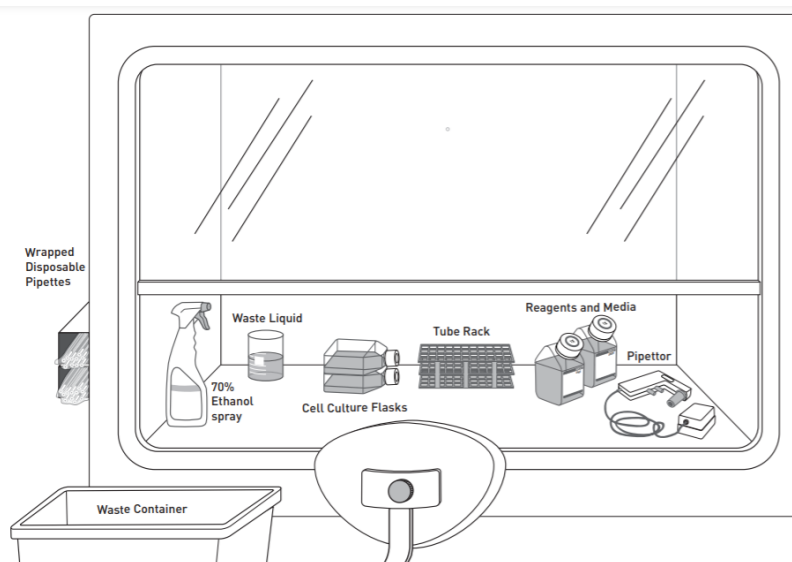


Figure 51: Biosafety Cabinet with cell culture equipment [10]

Correct amounts of reagents and cell media are pipetted, along with the cells, into a cell culture flask to dilute them appropriately before being placed in the incubator overnight.



Figure 52: Cell culturing in the Biosafety Cabinet

The cell culture flask is monitored under a microscope for over a week to check the cell growth, and cell media is changed when needed to replenish nutrients for cells to continue growing. The standard for monitoring cell growth is called confluency, which measures the percentage of the surface area covered with cells when observed in two-dimensional culture [11]. When the cells are observed to fully cover the cell culture flask when viewed under the microscope,



it would mean that it has reached 100% confluency and would be ready for harvesting and usage.



Figure 53: Cell culture reaching 20% confluency

Interestingly, I also learned how to request quotations during my internship. Even though this technical skill isn't related to my course of study, Biomedical Engineering, I believe it is still a critical skill to have when entering the workforce. Asking for reasonable quotations and comparing them would be beneficial to get the best quality for the most value.

I had also built upon many hard skills I had learnt in Temasek Polytechnic. I could apply my skills in soldering connections to be used for experiments and 3D design and printing. The 3D printer I had used uses resin as the printing material. However, it is very different from the resin printers I used in Temasek Polytechnic and requires a thorough cleaning procedure and alternate post-processing steps that include IPA and the ultrasonic cleaner.

The main project I had worked on was the Cartridge Alignment Program. Therefore, I gained much programming knowledge that builds upon what I learned in Temasek Polytechnic. I had learnt of Python's expansive collections of libraries that can be used for many different purposes, as well as serial communication with the components of the POC medical device.

Throughout the internship, I had also used Microsoft Office applications for many different applications. Company reports and write-ups were produced with Microsoft Word, data calculations and analysis with Microsoft Excel, and presentation slides were done with Microsoft PowerPoint. Therefore, I was familiar with their functions and learned new ones, such as playing multiple videos in a PowerPoint presentation.



### 4.3 REFLECTIONS ON STUDENT'S STRENGTHS AND WEAKNESSES

During the start of my internship, I was very unfamiliar with the microfluidic industry, despite coming from the Biomedical Engineering course. However, after reading up on some literature papers and learning through the experiments done, I overcame this weakness and was well-versed with the applications of microfluidics.

Another weakness I faced during this internship was not keeping track of some tasks. This had only occurred once, but it was a busy day where many tasks needed to be completed. I had left some cartridges in the oven over the weekend when they were only supposed to be in for an hour. This resulted in the need to replace the damaged backing film. From then onwards, I would always try to pay attention to the tasks at hand and set reminders on my phone if necessary.

One strength I have was that I was proficient in programming to produce the Cartridge Alignment Program. Even though I believe my co-workers would have been able to do the same, I am happy that I was able to help with this time-consuming task and thus allow more time for important tasks to be done, increasing the overall productivity of the team. I was also fortunate enough for the company to be willing to extend my internship indefinitely until I enlist for National Service as there are plans on expanding the program to replace other human error-prone processes, which would further improve the team's productivity. Therefore, I am glad to have worked on this project, and making it versatile enough to be integrated and used to improve the company's efficiency.

Another strength I have is being a fast learner. This is because I was able to pick up on the many different procedures that are done in the laboratory. It would only take me one or two tries to grab hold of the experiments, even with difficult tasks such as PDMS punching and PDMS device fabrication.

#### 4.4 REFLECTIONS ON STUDENT'S CAREER CHOICE AND DEVELOPMENT

After experiencing the Research and Development industry and learning more about biomedical engineering and biomedical sciences, I am now looking towards these options to pursue my further studies. Computer sciences have also piqued my interest as a potential career path, but I will take some time to consider which of the courses I am most interested in. I am also inspired by my supervisor's switch in her polytechnic diploma to a different engineering specialisation in university, and will consider my interests before making my decision. Researching has been fascinating to me as there is always something new to try every day, and it does not seem monotonous as a career path. Thus, I have applied to SUTD Honours And Research Programme (SHARP) and hopefully continue my education there. Even so, I will keep a look out for other university degree courses in NUS and NTU when they begin to take in applications.

#### 5. FEEDBACK AND SUGGESTIONS

The internship has been an eye-opening experience, and I have gained much from it. Both Temasek Polytechnic and CellWave Technologies have cared for me and never backed away when I felt unsure of anything. They had helped expose me to the Research and Development industry, which was the one I wanted to learn more about since many years prior. I am also grateful for CellWave Technologies for offering me an extension of my internship to continue working with them and expand on the Imaging and Alignment projects.

However, one small suggestion I have for Temasek Polytechnic is to have the deadlines and final presentation date to be further from the public holidays such as Chinese New Year. Even though I was fine with this arrangement, I believe this has caused inconvenience to some of my course mates. Other than that, I am glad that the internship went smoothly, and I truly appreciate everyone that has been with me throughout this fruitful internship journey.

Word Count (min 3000): 7409

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## APPENDIX

### APPENDIX A: Equipment in SUTD BEATs Laboratory



Figure 1: Olympus CKX53 with U-HGLGPS Fluorescence light module



Figure 2: Laboratory fume hoods



Figure 3: Laboratory incubators for cell culture



Figure 4: Laboratory freezer



Figure 5: Ultra-low temperature freezer



Figure 6: Liquid nitrogen storage





Figure 7: Flow cytometer



Figure 8: Syringe pump



Figure 9: Syringe and filter unit





Figure 10: Weighing scale

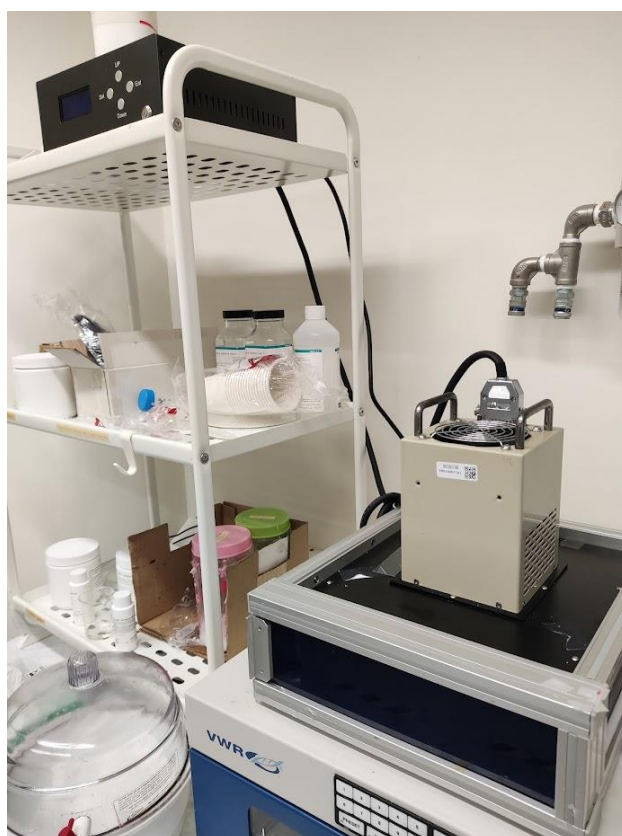


Figure 11: UV generator controls (top left) and chamber (bottom right)

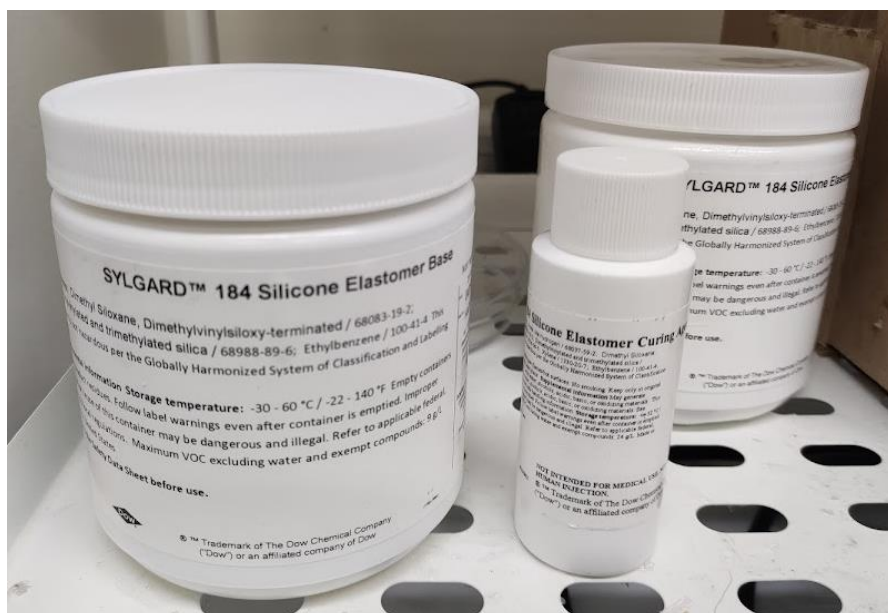


Figure 12: PDMS raw materials

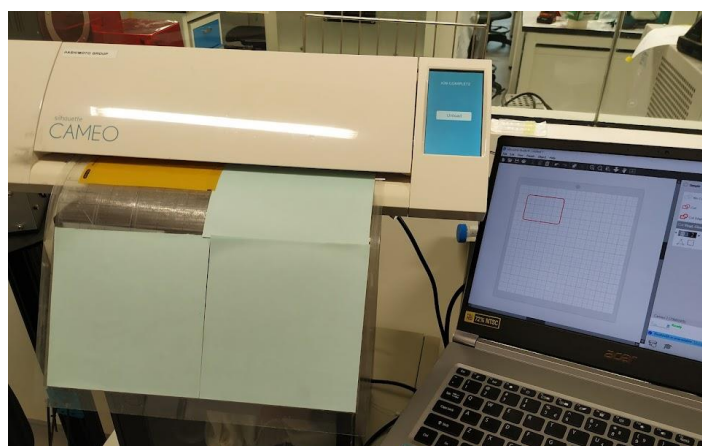


Figure 13: Cameo desktop cutter



Figure 14: Resin based 3D printer

## APPENDIX B: Experiments and Projects

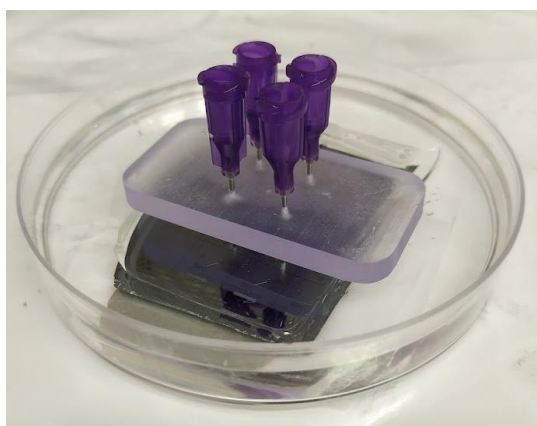


Figure 15: Forming holes within PDMS test

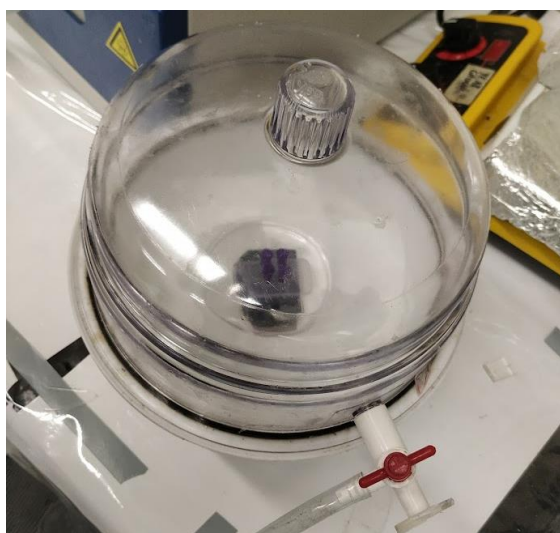


Figure 16: Forming holes setup degassing in vacuum chamber

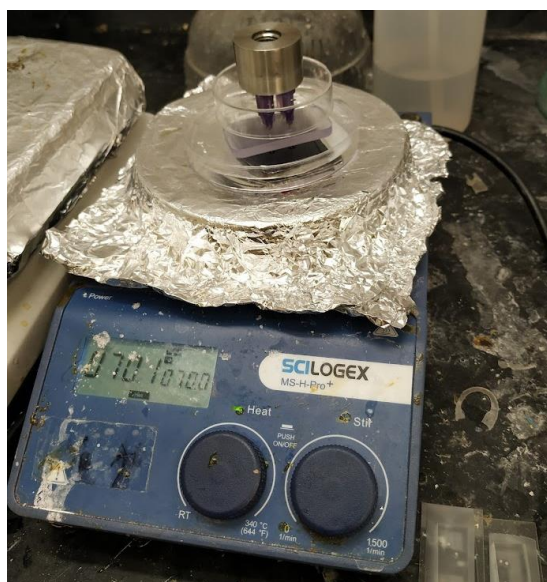


Figure 17: Forming holes setup on the hot plate



Figure 18: Inspection of formed holes



## APPENDIX C: Quality Control Checks

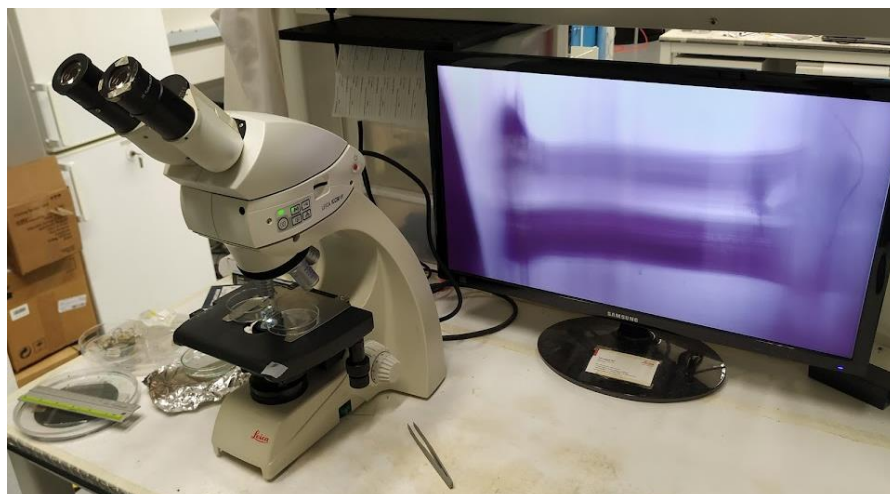


Figure 19: Quality Control of PDMS punched hole

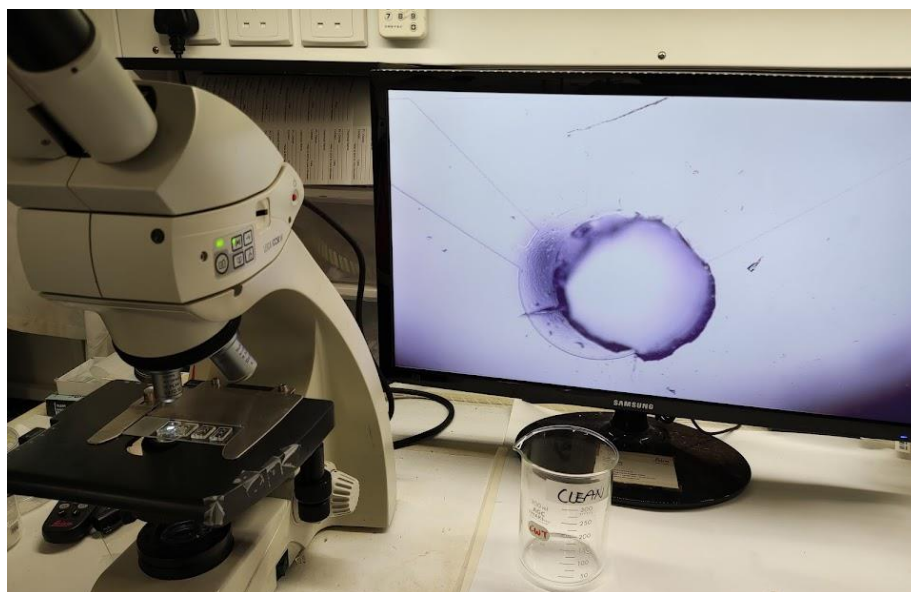


Figure 20: Quality Control of punched hole accuracy

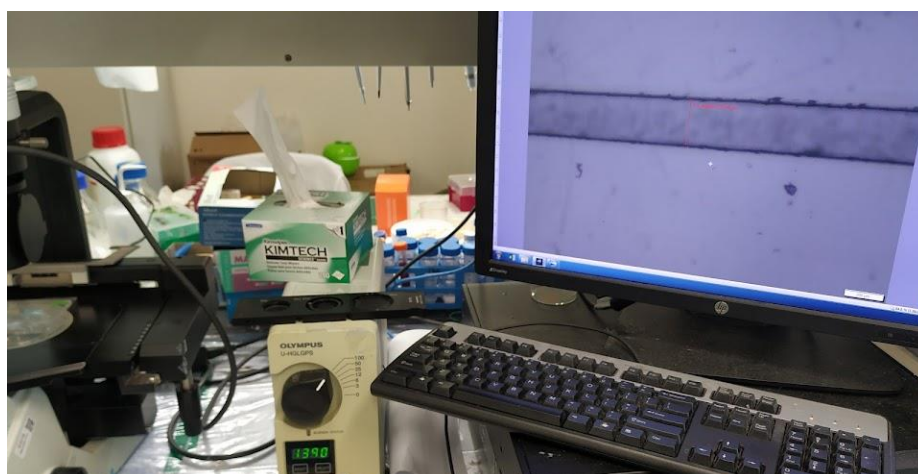


Figure 21: Quality Control of cartridge microfluidic channel dimensions

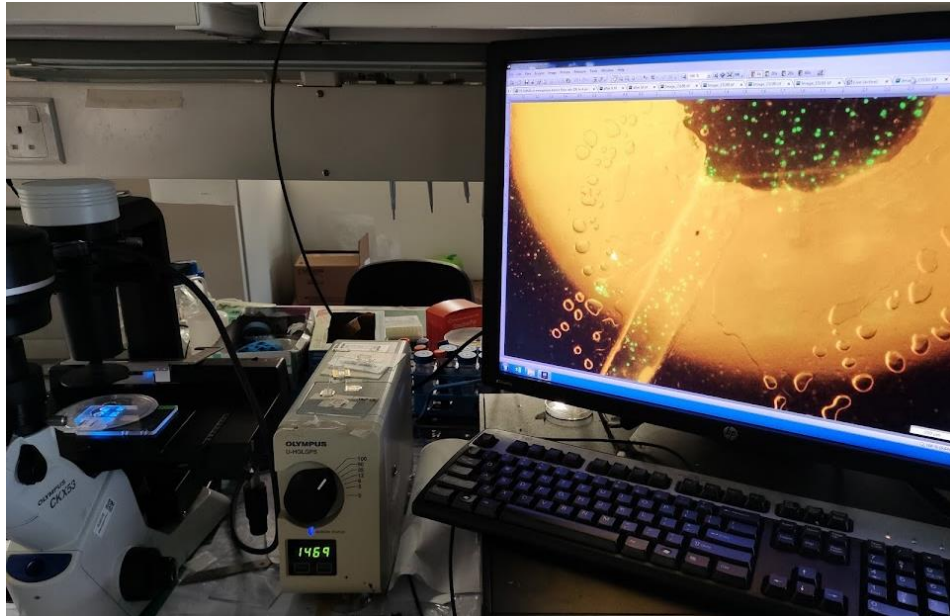


Figure 22: Post experiment inspection under fluorescence



Figure 23: Quality Control of cartridge debris